

From Hour-Long Drips to a One-Minute Jab: How a New Cancer Shot Is Freeing Up Patients' Lives

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For years, cancer patients receiving the blockbuster drug Keytruda have faced a tedious routine: sitting in a hospital chair for over an hour while the medicine drips slowly into their vein. But a new injectable version of the **immunotherapy**, now being rolled out by NHS England, promises to shrink that ordeal to just a couple of minutes. Shirley Xerxes, an 86-year-old from St Albans, was among the first to experience the difference at Mount Vernon Cancer Centre. "I was in the chair for a matter of minutes instead of an hour or more," she said. "It's made such a difference and gives me more time to live my life, including spending more time gardening." Keytruda, also known as **pembrolizumab**, is a type of **immunotherapy** that helps the body's own immune system recognize and attack cancer cells. Since 2015, NHS patients have received it via **intravenous infusion** a process that requires hospital pharmacy teams to prepare the drug under sterile conditions and administer it through a **cannula**. The new **formulation**, however, is a simple injection that can be given in as little as one minute every three weeks, or two minutes every six weeks, depending on the diagnosis. Around 14,000 patients in England start a course of Keytruda each year, and most are expected to transition to the injectable version. The drug is currently approved for 14 types of cancer, including lung, head and neck, cervical, and breast cancers. This shift is not just about patient comfort; it's a logistical game-changer. "Managing cancer treatment and regular hospital trips can be really exhausting," said an NHS spokesperson. "If we can do this in a much shorter period of time, it frees up space in our chemotherapy units and means we can start thinking about giving treatment in the community and away from hospitals." Keytruda works by blocking a "stop signal" that cancer cells use to hide from the immune system a mechanism often described as an "invisibility cloak." The discovery of this **immune checkpoint pathway** earned James Allison and Tasuku Honjo the 2018 Nobel Prize in Medicine. Keytruda was one of the earliest drugs to exploit this breakthrough, and it has since become the world's best-selling prescription medicine, with global sales of \$30 billion in 2025. The new jab is the third **immunotherapy** of its kind to be offered via injection on the NHS, following a similar reformulation of Opdivo (nivolumab) last year. For patients like Xerxes, the benefit is measured in moments reclaimed. "It's made such a difference," she said. And for the NHS, it's a step toward more efficient, patient-centered care.

Word Treasure

immunotherapy -- A treatment that helps the body's own immune system fight diseases like cancer.

pembrolizumab -- The scientific name for the cancer drug Keytruda.

intravenous infusion -- A way to give medicine directly into a vein through a tube, often taking a long time.

cannula -- A thin tube inserted into a vein to deliver medicine or fluids.

immune checkpoint pathway -- A natural stop signal that can be misused by cancer cells to hide from immune attacks.

formulation -- The specific way a drug is prepared, such as a liquid for injection or an intravenous drip.

Background you need

Cancer immunotherapy has been a major breakthrough in medicine, teaching the immune system to recognise and destroy tumours. The 2018 Nobel Prize in Physiology or Medicine was awarded to James Allison (an American immunologist) and Tasuku Honjo (a Japanese immunologist) for discovering how to release molecular 'brakes' on immune cells--a mechanism known as immune checkpoint inhibition.

Keytruda, developed by the pharmaceutical company Merck, was first approved by the U.S. Food and Drug Administration in 2014 and quickly became one of the world's best-selling drugs, generating \$30 billion in 2025 alone. For years, patients needed to visit hospital pharmacies where the medicine was mixed under sterile conditions and infused slowly through a cannula, tying up both patient time and hospital resources.

The new injectable formulation simplifies this dramatically, much like a routine vaccination. NHS England (the government body that runs the health service in England) is rolling it out to free up chemotherapy unit space and eventually offer treatment in community clinics. This shift follows a similar reformulation of another immunotherapy, Opdivo (nivolumab), which the NHS adopted last year.

The underlying science--blocking the 'invisibility cloak' that cancers use to evade the immune system--has transformed survival rates for many cancers, including lung, head and neck, cervical and breast cancers. Faster administration not only eases pressure on hospitals but also gives patients precious extra time to spend at home.

Break it down

LEAD: Personal story - 86-year-old Shirley Xerxes describes how the new injection at Mount Vernon Cancer Centre saved her hours.

WHAT & HOW: Explanation of Keytruda (pembrolizumab) and its role as an immunotherapy.

+ OLD vs NEW: Intravenous infusion (over an hour, sterile prep, cannula) vs simple injection (one to two minutes every few weeks).

SCOPE & BENEFITS: Around 14,000 patients per year, 14 approved cancers; frees up chemotherapy units, enables community treatment.

MECHANISM: Blocks the immune checkpoint 'stop signal' that cancer cells use as an invisibility cloak.

+ NOBEL CONTEXT: Discovery of the immune checkpoint pathway by Allison and Honjo, awarded the 2018 Nobel Prize.

PRECEDENT & SCALE: Third NHS immunotherapy injection, following Opdivo; Keytruda's global sales of \$30 billion in 2025.

CLOSING: Return to patient Xerxes, emphasising reclaimed time and NHS efficiency.

Why it matters

For a teenager, this story shows how medical advances can directly improve the quality of life for patients--freeing up time for family, hobbies and rest, while also making healthcare systems more efficient. If a family member ever needs cancer treatment, innovations like these could mean fewer long days in hospital and more treatment closer to home.

Test what you remember

1. What is the generic name of the drug Keytruda?
 - (A) Nivolumab
 - (B) Pembrolizumab
 - (C) Opdivo
 - (D) Rituximab
2. How quickly can the new injection version of Keytruda be given?
 - (A) About 5 minutes every day
 - (B) As little as one minute every three weeks
 - (C) Around 10 minutes once a month
 - (D) One hour every six weeks
3. What does Keytruda block to help the immune system attack cancer?
 - (A) A growth hormone
 - (B) A stop signal used by cancer cells
 - (C) Blood supply to tumors
 - (D) Pain receptors
4. Who won the 2018 Nobel Prize in Medicine for discovering the immune checkpoint pathway?
 - (A) James Watson and Francis Crick
 - (B) Rosalind Franklin and Maurice Wilkins
 - (C) James Allison and Tasuku Honjo
 - (D) Marie Curie and Pierre Curie
5. Approximately how many patients in England start a course of Keytruda each year?
 - (A) 1,400
 - (B) 14,000
 - (C) 140,000
 - (D) 1,400,000
6. The new Keytruda injection is the third immunotherapy of its kind on the NHS. Which similar drug was offered as an injection before it?
 - (A) Herceptin
 - (B) Opdivo (nivolumab)
 - (C) Tamoxifen
 - (D) Avastin

Answer key (try the quiz first!): 1: B 2: B 3: B 4: C 5: B 6: B

Questions to think about

- 1. Positive - patient convenience and system gains:** Patients gain back hours previously spent in hospital chairs, making treatment less disruptive. The NHS can use the saved chair and pharmacy time to treat more people, reduce waiting lists and eventually offer injections in local clinics, bringing care nearer to patients' homes.
- 2. Critical - risks of reduced monitoring:** Not all patients can safely switch to the injectable form, and some may have reactions at the injection site. Moving care away from hospitals could mean fewer chances for staff to spot complications early, raising concerns about safety if community centres lack the same level of immediate support.
- 3. Neutral - a trend in treatment delivery:** The shift from intravenous infusions to quick injections mirrors a broader trend in medicine: turning hospital-only treatments into outpatient or home-based procedures. It reflects how drug formulation advances and healthcare logistics evolve together, decoupling effective treatment from the need for a hospital bed.
- 4. Forward-looking - the future of cancer care:** If injections can be given safely by nurses in the community or even at home, cancer treatment could one day become as routine as managing a chronic condition like diabetes. This could transform cancer from a disease that dominates life to one that is managed alongside daily activities, greatly reducing its emotional and economic burden.

MY THOUGHTS (at least 20 words):